ALGEBRA 2 · UNIT 2: LINEAR FUNCTIONS & SYSTEMS

Farm Planner

Use linear programming and systems in three variables to run a farm.

Honors

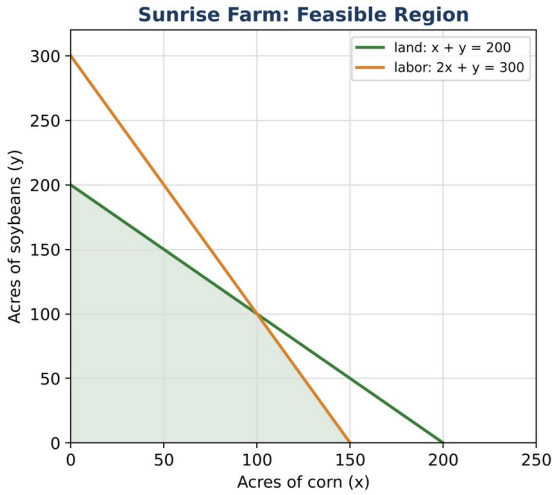
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Sunrise Farm grows corn and soybeans. Land and labor are limited, and each crop brings a different profit. The farmer wants the plan that earns the most money.

This is linear programming: graph the constraints, find the corners of the feasible region, and test them. Then you will mix feed for the farm's animals with a three-variable system.



Per Acre

Corn: 2 h labor

profit $300

Soybeans: 1 h labor

profit $200

Land ≤ 200 acres

Labor ≤ 300 h

Learning Goals

- I can solve linear programming problems with several constraints.
- I can explain multiple optimal solutions.
- I can solve and interpret three-variable systems.

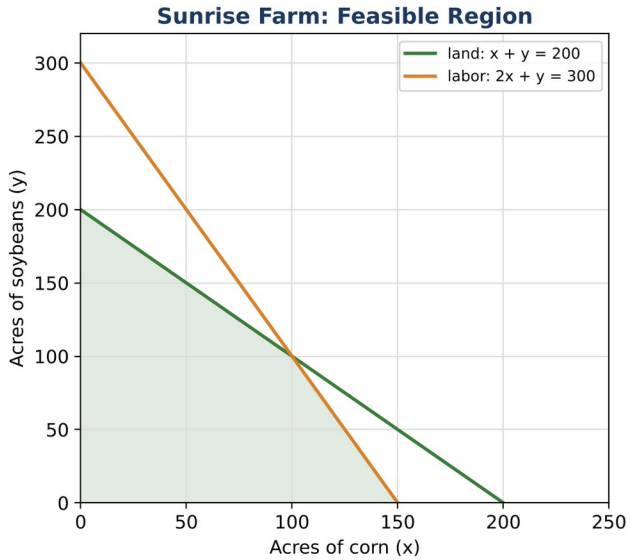
Project Timeline

When	What you will do
Day 1	Constraints & Vertices
Day 2	Optimize & Analyze
Day 3	Mix the Feed
Final	Farm Business Plan + reflection

Part 1: Constraints & Vertices

Day 1 · Add a budget: corn costs $150 per acre and soybeans $100 per acre; budget $24,000.

1. Write all constraints and graph them.



Per Acre
Corn: 2 h labor
profit $300
Soybeans: 1 h labor
profit $200
Land ≤ 200 acres
Labor ≤ 300 h

2. Find every vertex of the new feasible region.

3. Which constraints are binding at each vertex?

Part 2: Optimize & Analyze

Day 2 · Profit P = 300x + 200y.

4. Evaluate P at each vertex. What do you notice?

5. Explain the tie using slopes. List three different optimal plans.

6. For what corn profit k (with soybeans still $200) would (120, 60) be the only best plan?

Part 3: Mix the Feed

Day 3 · Feed uses corn (a), soy meal (b), and oats (c).

7. Solve: a + b + c = 100; 0.09a + 0.44b + 0.12c = 16.6; 0.10a + 0.30b + 0.15c = 15.

8. If the protein target changes to p lb (cost and total stay the same), write a and b in terms of p (with c also in terms of p).

9. Find the range of p that keeps every amount nonnegative.

Final Product: Farm Business Plan

Your final product must include:

- Graph with all vertices and binding constraints
- Tie explanation and sensitivity analysis
- Feed solution and parametric analysis
- Business plan

Reflection

R1. What does it mean for a farmer that many plans tie?

R2. Why might a real feed problem have a range of good answers?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked. Goes beyond the minimum with a creative, realistic choice.	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Polished, organized, and visually engaging.	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.